IMPERIAL

Economic evaluation of a new treatment for kidney small masses

Economic and Clinical Evaluation of Cryoablation and Partial Nephrectomy in Managing Small Renal Masses

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Abstract

Background

Small renal masses (SRMs) are increasingly detected due to advanced imaging, yet challenges persist in diagnosing malignancy pre-treatment. Current management includes active surveillance (AS), partial nephrectomy (PN), radical nephrectomy (NRL) and cryoablation. While PN remains the gold standard, percutaneous cryoablation (PCA) offers a minimally invasive alternative. However, the cost-effectiveness of these interventions and the role of biopsy in guiding treatment decisions, even though it could potentially guide treatment decisions by distinguishing malignant from benign lesions, remain understudied.

Aims

(1) Compare the cost-effectiveness and clinical outcomes of multiple treatments for SRMs.

(2) Evaluate whether incorporating biopsy prior to treatment decision-making improves overall cost-effectiveness by reducing unnecessary interventions.

Methods

A cost-utility analysis was conducted using data from a UK-based RCT (n=200) comparing PCA (n=100) and RAPN (n=100). Decision-tree and Markov models were employed to assess short- and long-term (20-year) outcomes, respectively. Costs were derived from NHS tariffs (2024–2025), and QALYs were calculated using EQ-5D-5L data. Incremental Cost-Effectiveness Ratios (ICERs) and Net Monetary Benefits (NMBs) were computed at £20,000/QALY and £30,000/QALY thresholds. Sensitivity analyses tested biopsy cost variability (£600–£1,800) and AI-driven utility improvements.

Results

RAPN yielded marginally higher QALYs (0.7467 vs. 0.7244/patient) than PCA, but PCA had superior NMBs (£1,492,482.95 vs. £987,272.96 at £30,000/QALY). ICERs favored RAPN, as PCA exceeded NICE's threshold. Biopsy costs significantly influenced outcomes: AI-assisted biopsy (10% utility gain) reduced ICERs to £25,639.78, making it cost-effective. The biopsy-driven strategy dominated long-term modelling, with higher QALYs (3,482 vs. 2,959) and lower costs (£6.07M vs. £7.90M) versus no-biopsy. The annual budget impact was £3.01 million for an 8-million population.

Conclusion

PCA is economically favorable in specific contexts (e.g., AI-enhanced biopsy), though RAPN offers slight clinical advantages. A biopsy-first strategy reduces unnecessary surgeries and long-term costs, but its feasibility depends on regional pricing and technology adoption. These findings support reconsidering the role of biopsy in SRM management, particularly as diagnostic technologies continue to advance.

Keywords

Renal cancer, small kidney masses, cryoablation, nephrectomy, biopsy, cost-effectiveness analysis

0 Introduction

Kidney cancer is the 14th most commonly diagnosed cancer globally with 434,840 new cases reported in 2022. (1) Small renal masses (SRMs) are kidney tumours with typically ≤4 cm in diameter. Approximately 80% of SRMs are malignant while 20% are benign lesions like oncocytomas or angiomyolipomas.(2) SRMs are detected incidentally due to the widespread use of advanced imaging techniques. However, despite improvements in imaging techniques and the increasing use of renal mass biopsy accurately diagnosing malignancy before treatment remains a challenge. (3)

Treatment options range from active surveillance to surgery depending on factors like tumor characteristics and patient health.

1. Active Surveillance - This should be a primary management option for patients with significant comorbidities and limited life expectancy. (4)
2. Partial Nephrectomy - Partial nephrectomy (PN) for SRMs is the standard treatment that should be offered to all patients for whom an intervention is indicated and who possess a tumor that is amenable to this approach. (5)
3. Cryoablation (percutaneous) - It includes cryosurgical, microwave and radiofrequency techniques offering a minimally invasive approach that uses heat or freezing to destroy tumors. Suitable for patients where complete ablation is achievable. (3)
4. Radical nephrectomy – Should be considered only for patients with highly complex tumors that are not suitable for partial nephrectomy (PN) or when PN poses an unacceptable risk. Referral to a surgeon and a center with PN expertise is recommended. (6)

1 Aims

The aims of the study are-

1. Comparing the cost-effectiveness and clinical outcomes of percutaneous cryoablation and robotic-assisted partial nephrectomy for small renal masses (SRMs).
2. Conducting a "within trial" evaluation using data from a randomized clinical trial involving 200 patients to compare cryoablation and partial nephrectomy
3. Constructing decision trees to assess whether performing a biopsy before surgery (cryotherapy or nephrectomy) is cost-effective using trial data and literature

2 Method

We conducted a cost-utility analysis to compare the costs and outcomes of treatments for small renal masses using data from a randomized controlled trial (RCT) and existing literature. The principal RCT conducted in London, United Kingdom included 200 adult patients diagnosed with small renal masses. Participants were randomly assigned to one of two equal groups: 100 patients received percutaneous cryoablation while the other 100 underwent robotic-assisted partial nephrectomy. This randomized design ensured a systematic comparison of both treatment modalities. The evaluation was conducted from the perspective of the UK National Health Service (NHS) in alignment with NICE guidelines for health technology assessment. This perspective captures direct medical costs relevant to public healthcare decision-making in the UK.

The study prospectively tracked clinical outcomes and comprehensive cost metrics for both groups. The primary outcome measure was quality-adjusted life years (QALYs) which integrate life expectancy and quality of life to facilitate direct comparison between the two interventions. QALYs are a standard metric in health economic evaluations enabling comparisons between treatments that differ in both survival rates and patient-reported quality of life. This makes them particularly useful for assessing treatments for small renal masses where long-term quality of life may vary depending on the intervention.

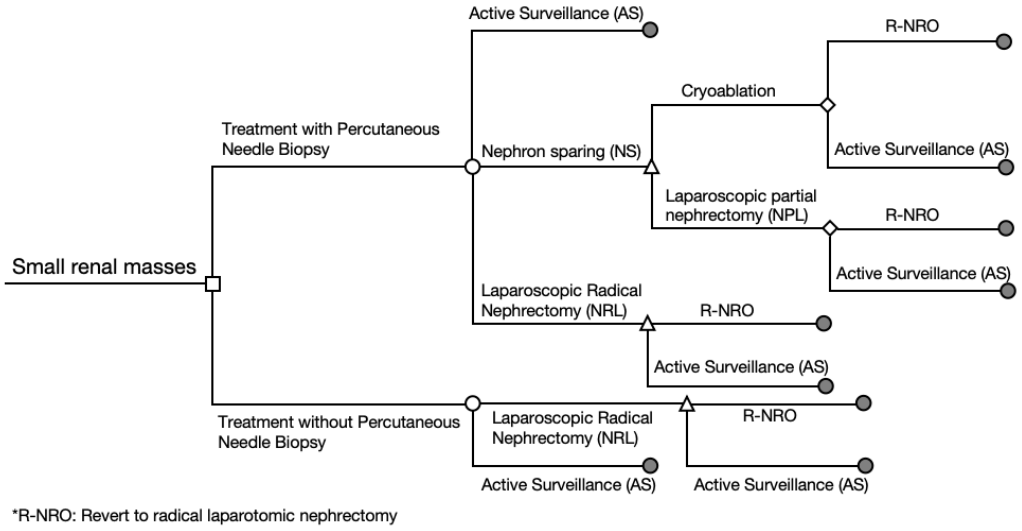
2.1 Model structure

We constructed a decision tree to evaluate two distinct intervention pathways for the short-run decision analysis: one utilizing biopsy to inform the decision-making process for cryoablation and the other proceeding directly to nephrectomy without the preceding biopsy. In the no biopsy group, patients follow a pathway where they may be eligible for radical nephrectomy or placed directly on active surveillance (AS). When deemed eligible for intervention, successful procedures lead to AS, while complications may necessitate reverting to a laparotomic radical nephrectomy (R-NRO).

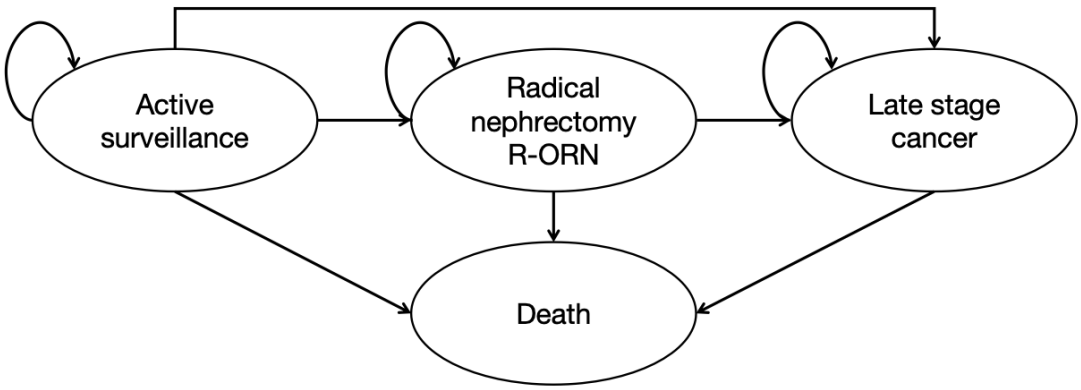
Conversely, conducting a pre-treatment biopsy introduces additional decision points: patients may be determined suitable for active surveillance without further intervention or identified as requiring more definitive treatment. Potential treatment options include laparoscopic radical nephrectomy or nephron-sparing procedures that preserve kidney function. Like the no-biopsy pathway, both interventional approaches (laparoscopic partial nephrectomy and cryoablation) carry the potential risk of complications that could ultimately result in a laparotomic radical nephrectomy.

In our long-term model, we employed a Markov model to track patient trajectories over a 20-year period following initial treatment decisions with or without biopsy. From the decision tree, patients will transition between primary health states: active surveillance and laparotomic radical nephrectomy (R-ORN) as the initial health states of the Markov model. Patients then progress from initial health states to late-stage cancer or mortality, guided by established transition probabilities. This model captures the long-term clinical progression and potential outcomes associated with various initial treatment approaches. This time horizon was selected to reflect the average life

expectancy of the target population and to ensure that the long-term costs and benefits associated with early intervention or surveillance were fully captured.



(a)



(b)

Figure 1. (a) Decision Tree model (b) Long-run Markov model

2.2 Cost

The costs associated with biopsy, diagnosis, and surgery can be found in the NHS Pricing Schedule for 2024-2025. (The HRG code for the Percutaneous Needle Biopsy of a Kidney Lesion for patients aged 19 years and over is YL20A, and the price is £778. Staff costs are based on data from the Personal Social Services Research Unit (PSSRU) Unit Costs of Health and Social Care 2023/2024. While most drug costs are available on the BNF website, the price for chlorhexidine acetate 0.02% irrigation solution (1 litre bottle containing 200 micrograms per 1 ml) was not listed. Therefore, we sourced this price from Baxter Healthcare's website.)

To ensure all costs were comparable in real terms, historical expenditures are adjusted to 2024 price levels using annual Consumer Price Index including Housing (CPIH) inflation rates published by the Office for National Statistics. (7) The adjustment process applied a cumulative inflation multiplier to each cost item based on its original year. For a cost incurred in year t, its 2024-equivalent value was calculated as:

$$\text{Adjusted Cost}_{2024} = \text{Original Cost}_t \times \prod_{i=t}^{2023}(1 + \text{Inflation Rate}_i)$$

where Inflation Rate corresponds to the CPIH annual percentage change for year *i*. Notably, costs sourced from recent years (e.g., 2023 NHS Pricing Schedule) required minimal adjustment, whereas earlier figures (e.g., 2005 biopsy prices) demanded compounding over two decades of variable inflation.

According to HRG tariff, the cost of biopsy exhibits noticeable variation ranging from £600 to £1800. With the highest cost of £1,800, the biopsy could be guided and performed with innovative Artificial Intelligence technology. Variations in the cost of biopsy procedures can significantly influence the overall treatment costs within clinical trials thereby affecting subsequent decision tree analyses, cost-effectiveness analyses (CEA), and sensitivity assessments. Since biopsy expenses directly contribute to the total intervention costs, fluctuations in pricing may alter the ICER values and cost-effectiveness conclusions of comparative pathways. In decision tree modelling, such cost variability could shift thresholds for preferred strategies. For ICER analysis, higher biopsy costs may reduce the economic attractiveness of biopsy-guided approaches if the additional diagnostic expense is not offset by superior clinical outcomes. Conversely, lower biopsy costs could make these strategies more favorable in CEA by improving their cost-effectiveness profiles.

Sensitivity analyses must explicitly account for the uncertainty surrounding biopsy costs, as these costs serve as a critical parameter that can significantly influence the stability of model outcomes and the reliability of policy recommendations.(8) Variability in biopsy costs driven by factors such as procedural complexity, healthcare setting and regional pricing differences may lead to substantial fluctuations in the overall cost-effectiveness estimates of the interventions. Failing to address this uncertainty could undermine the credibility of the analysis, potentially leading to suboptimal policy decisions. Therefore, conducting scenario analyses or probabilistic sensitivity analyses that incorporate a range of plausible biopsy cost estimates is essential to ensure the robustness and generalizability of the findings.

2.3 Outcomes

The QALYs for both the cryoablation and robotic-assisted partial nephrectomy groups are measured at baseline, 4 weeks, and 12 months, utilizing available EQ-5D-5L data. The EQ-5D is a concise, generic measure that assigns a single summary value to each health profile. This measure has been instrumental in generating a series of health utility, facilitating the assessment of health

outcomes. These values, organized in country-specific sets, are used to determine QALYs. The value set is employed to calculate QALYs, with the weights reflecting the significance of each health profile. QALYs are anchored at 0 representing death and 1 indicating full health. The EQ-5D-5L encompasses 3,125 unique health profiles. (9)

QALYs offer a standardized way to assess the benefits of different interventions in terms of health-related quality of life and patient survival. The duration spent in a particular health state is adjusted based on the utility score assigned to that state. When the costs of these interventions are factored in, cost-utility ratios can be determined. These ratios show the additional expenses required to achieve a year of optimal health, equivalent to one QALY. (10)

2.4 Probabilities

In the Markov model, the transition probabilities for each health state are for both the control and intervention groups. In the control scenario, an individual in the AS state has a 50% chance of remaining in AS, a 30% chance of progressing to R-ORN, a 10% chance of developing late-stage cancer, and a 10% probability of death. In contrast, under the intervention, the probability of remaining in AS increases to 60%, accompanied by a reduced likelihood of progressing to R-ORN and late-stage cancer.

2.5 Measuring Cost-effectiveness

The cost-effectiveness analysis will be conducted using the ICER, which compares the difference in costs and QALYs between interventions. Both a trial-based ICER analysis and a Markov model will be employed to evaluate cost-effectiveness under different scenarios. Additionally, the Net Monetary Benefit (NMB) will be calculated using UK willingness-to-pay (WTP) thresholds of £20,000 and £30,000 per QALY. Results will be visualised through a cost-effectiveness plane (CEP) graph providing insights into the relative value of the interventions.

2.6 Sensitivity Analysis

Sensitivity analysis was conducted to assess the robustness of the results by examining the impact of variations in utility parameter estimates and associated costs.

Specifically, five one-way sensitivity analyses (SA1–SA5) were carried out. These included testing the effect of using the minimum and maximum costs for biopsy (SA1 and SA2), as well as evaluating the impact of a 10% increase in utility due to AI-supported care (SA3–SA5). SA3 assessed a combined scenario where the cost of biopsy was set to its maximum and utility was increased by 10%, SA4 examined the effect of increasing the cost of Active Surveillance, and SA5 evaluated the most favorable scenario with minimum biopsy cost and AI-enhanced utility.

These analyses allowed us to identify the key parameters influencing cost-effectiveness and examine whether biopsy remains a preferred strategy under different hypothetical conditions.

3. Results

3.1 Cost-effectiveness

The total QALYs for the cryoablation group (Group A) is 72.44 while the surgery group (Group B) achieves a total QALY of 74.67 with both groups comprising 100 participants. This results in an average QALY per patient of 0.7244 for Group A and 0.7467 for Group B. The findings indicate that, on average, patients undergoing surgery experience a slightly higher quality-adjusted life expectancy compared to those receiving cryoablation. The difference of 0.0223 QALYs per patient suggests a marginal advantage associated with surgical intervention.

However, QALY alone is not sufficient to determine the preferred intervention, as cost considerations and overall cost-effectiveness play a crucial role in decision-making. Therefore, we must analyze the cost differences between the two treatments and evaluate their ICER to assess the value of each intervention comprehensively since it compares the cost-effectiveness of two treatments by calculating the additional cost per QALY gained. The results obtained from our ICER trial are summarized as follows:

$$ICER = \frac{Cost_A - Cost_B}{QALY_A - QALY_B}$$

$$ICER = \frac{£680,609.55 - £1,252,955.79}{72.44 - 74.67} = £255,754.34$$

According to NICE, treatments with an ICER above £20,000–£30,000 per QALY are generally considered not cost-effective. Since cryoablation's ICER exceeds this threshold, it is deemed not cost-effective compared to surgery.

In contrast, the NMB considers both costs and health benefits together. It is calculated using the formula:

$$NMB = (\lambda \times \text{QALYs}) - \text{Costs}$$

where λ (lambda) represents the willingness-to-pay per QALY (e.g., £20,000 or £30,000). At the £20,000 threshold, cryoablation has an NMB of £768,118.79 compared to £240,530.04 for surgery. This difference becomes even more pronounced at the £30,000 threshold where cryoablation NMB reaches £1,492,482.95, while Surgery's stands at £987,272.96.

These findings indicate that cryoablation offers greater overall value making it the more cost-effective option at both thresholds.

3.2 Sensitivity Analysis

A series of five sensitivity analyses were undertaken to evaluate the model's cost and utility parameters. The analyses explored various scenarios, including the costs of biopsy assessed at both minimum and maximum levels, as well as the potential impact of employing artificial intelligence

(AI), resulting in a 10% increase in utility. Additionally, the analyses considered the scenario in which the cost of active surveillance was maximized, and finally minimizing the biopsy cost, with the assumption of a 10% enhancement in utility attributable to financial support from a charitable foundation. The findings from each sensitivity analysis were systematically summarized, including the ICER, QALY and NMB as presented in Table 1.

Table 1. Sensitivity Analysis Results

Sensitivity Analysis	Parameter	ICER	QALY	NMB
SA1	Minimum biopsy cost	£35,213.10	0.77	at £20,000: £7,249.06
SA2	Maximum biopsy cost	£50,152.02	0.77	at £20,000: £6,043.14
SA3	AI improves the utility of biopsy patients by 10%	£25,639.78	0.85	at £30,000: £16,028.82
SA4	Maximum active surveillance cost	£37,199.94	0.77	at £30,000: £14,471.86
SA5	AI improves the utility of biopsy patients by 10% and minimum biopsy cost	£18155.42	0.85	At £30,000: £1835.91

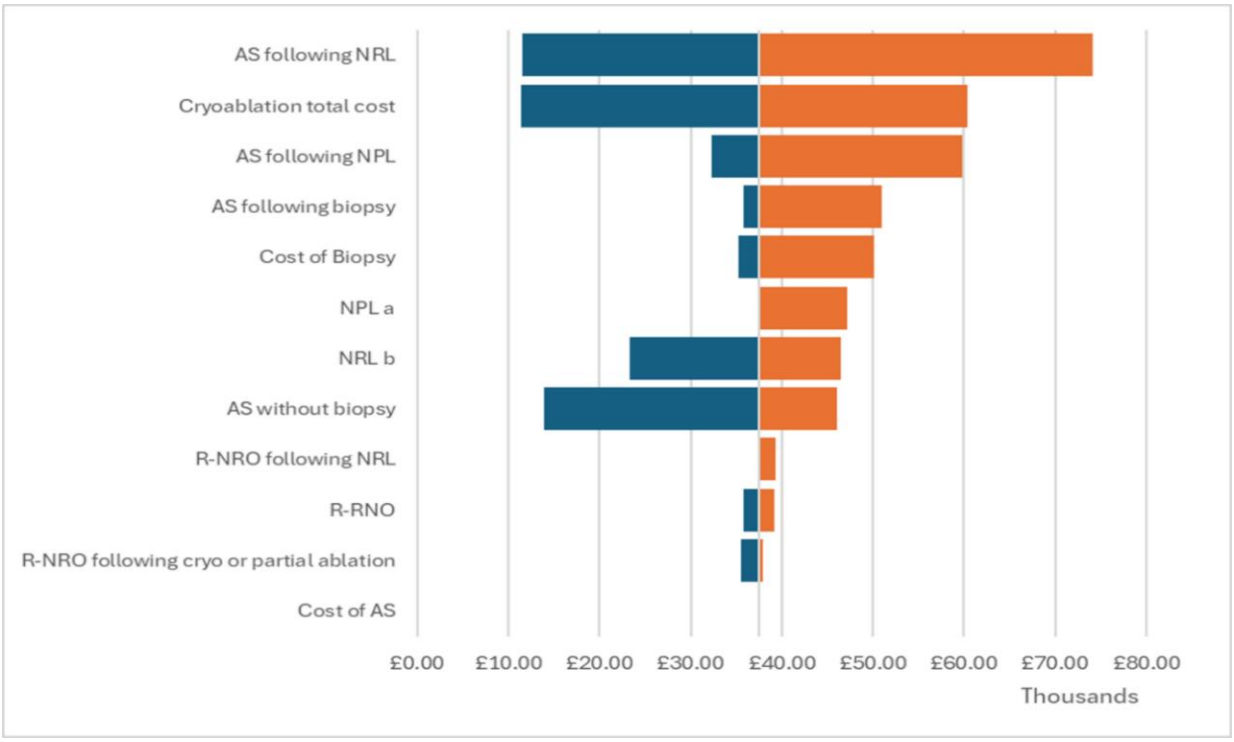


Figure 2. Sensitivity Analysis – Tornado diagram

SA1 tested the impact of reducing the cost of biopsy to its minimum value. Although this lowered overall costs, the ICER remained high (£35,213.10) and biopsy still did not surpass no-biopsy in terms of NMB, indicating limited impact from cost reduction alone.

SA2 explored the effect of increasing biopsy costs to their maximum level. This resulted in a notably higher ICER (£50,152.02) and substantially reduced NMBs at both willingness-to-pay thresholds. The findings show that higher biopsy costs significantly weaken its cost-effectiveness.

SA3 examined a combined scenario in which the cost of biopsy was set to its maximum and AI technology improved patient utility by 10%. Despite the higher cost (£9,321.18), the increase in QALYs (from 0.77 to 0.85) led to a notably lower ICER of £25,639.78, falling below the £30,000 willingness-to-pay threshold. This result suggests that improvements in patient outcomes can offset higher procedure costs, making biopsy a potentially cost-effective strategy under these conditions.

SA4 assessed the effect of increasing the cost of Active Surveillance to its maximum value. The results showed a modest increase in the ICER (£37,199.94), with little change in NMBs. Under the assumption of increased Active Surveillance costs, biopsy does not appear to be a more favorable option.

SA5 combined the two most favorable assumptions: a 10% increase in utility due to AI and the lowest possible biopsy cost enabled by charitable funding. This scenario produced the most favorable outcomes, with a significantly low ICER of £18,155.42 and the highest NMB at both £20,000 and £30,000 thresholds. These findings strongly support the potential cost-effectiveness of biopsy when clinical outcomes are improved, and economic barriers are minimized.

Overall, the 5 sensitivity analyses highlight that improving health outcomes (utility) has a greater impact on cost-effectiveness than altering costs alone and that combined strategies (like in SA5) may be the most effective path forward for making biopsy a cost-effective option in SRM management.

Minimizing biopsy costs leads to reduced expense; however, merely lowering the cost of biopsy does not ensure it becomes a cost-effective strategy. Conversely, significantly increasing the costs associated with biopsy greatly diminishes its cost-effectiveness. Assuming that AI enhances the utility of biopsy for patients by 10%, this improvement has a substantial positive effect, potentially rendering biopsy a cost-effective strategy. On the other hand, raising the costs of AS to their maximum level has a modest impact yet biopsy remains less cost-effective in comparison.

3.3 Budget Impact Analysis

The Budget Impact Analysis (BIA) evaluates the financial implications of implementing a biopsy strategy for patients with Small Renal Masses in an 8-million population, where the incidence rate of SRM is 7 per 100,000 individuals. The total biopsy cost per year is £348,544, while the total cryoablation cost per year amounts to £2,664,056.37 resulting in a total annual cost of

£3,012,600.37. In terms of short-term burden, the cost of implementing this strategy in the first year is £3.01 million, representing significant budget consideration. Over the long term, the financial burden increases linearly, potentially reaching a total of £15.06 million over five years if no cost reductions or efficiencies are introduced.

3.4 Markov Model

Each state in the Markov model is associated with a specific cost for late-stage cancer being the most expensive at £10,000 per patient while the costs for AS and R-ORN are significantly lower at £200 and £300 respectively. Utilities, which represent the QALYs for each state, indicate that AS has the highest utility value of 0.9 whereas late-stage cancer has a notably lower utility of 0.4. In terms of discounting and long-term considerations, a discount rate of 3.5% is applied in accordance with UK Green Book guidelines to account for the diminishing value of future costs and benefits. As detailed in the Green Book reference table, discount rates decrease over time decreasing to 3.0% after 30 years and further down to 1.0% beyond 300 years. Intervention decreases the likelihood of progressing to severe health states reducing long-term costs and enhancing the patient's quality of life. Although it may involve higher short-term expenses, it effectively delays disease progression which could result in cost savings when managing late-stage cancer cases.al short-term costs, it delays disease progression, potentially leading to cost savings in managing late-stage cancer cases.

After evaluating the cost and QALYs in the ICER Markov model with discounted utility and discounted cost, it is evident that the biopsy is the preferred approach as it incurs lower total costs (£6.07M vs. £7.90M) results in higher total QALYs (3,482 vs. 2,959) compared to non-biopsy and leads to more patients receiving R-ORN treatment. The ICER was substantially lower for the biopsy compared to the no-biopsy (£ 1741.64 vs. £ 2668.34) with an overall difference of £3501.81, thereby suggesting that early intervention may slow progression to late-stage disease while achieving better health outcomes and significant cost savings. Furthermore, the NMB analysis supports this conclusion. At the £ 2,000 threshold, the biopsy achieved an NMB of £ 63,582,263.76, which is £ 12,291,966.65 higher than no-biopsy. At the £ 3,000 threshold, this difference increased to £ 17,522,187.91.

4. Discussion

Our analysis provides a comprehensive economic evaluation of cryoablation and surgery, as well as the cost-effectiveness of a biopsy-led strategy for managing SRMs. The findings offer valuable insights for healthcare decision-makers highlighting the trade-offs between different treatment modalities and the potential economic benefits of early intervention and technological advancements. The discrepancy between the ICER and NMB results for cryoablation underscores the importance of considering multiple economic metrics and the sensitivity of these analyses to the chosen willingness-to-pay threshold. The sensitivity analyses for biopsy provide practical guidance on factors that could enhance its cost-effectiveness, such as integrating AI to improve

diagnostic accuracy and patient outcomes. The budget impact analysis highlights the financial implications of implementing a biopsy strategy which is crucial for resource planning. Finally, the Markov model provides a compelling argument for the long-term economic and clinical benefits of a biopsy-driven approach.

Although our analysis employs different methodologies, it builds upon the foundational insights provided by prior research in this rapidly evolving field. By addressing critical gaps and introducing novel perspectives, our findings contribute to a deeper understanding within the context of healthcare decision-making. A cost comparison study reveals that percutaneous cryoablation is significantly more cost-effective than both open and robot-assisted laparoscopic partial nephrectomy for similar renal tumors, with savings of $5,000 and $6,400 respectively. This cost advantage persists despite cryoablation being used in older, higher-risk patients, and is partly due to shorter hospital stays. However, the higher pathology costs associated with partial nephrectomies provide a more definitive diagnosis compared to cryoablation's core biopsy. While cryoablation shows clear financial benefits, treatment decisions should balance cost with oncological outcomes and patient factors. (11)

Another study revealed that partial nephrectomy (PN) and percutaneous cryoablation (PCA) provide similar outcomes for patients with cT1a renal cell carcinoma (RCC), with PCA yielding 3.68 QALYs compared to 3.67 QALYs for PN. Furthermore, PCA appears to be more cost-effective, with overall expected costs of $20,491 for PCA versus $26,478 for PN. (12)

The use of biopsy to guide treatment decisions allows clinicians to more accurately determine which patients genuinely require surgical intervention and which can be safely managed through alternative means. This approach not only has the potential to prevent unnecessary surgeries but also aligns with patient-centered care principles by reducing the risk of overtreatment. Biopsy can be considered as a cost-effective triage tool, showing promising results in terms of life expectancy outcomes, comparable to those of immediate surgery while offering significant cost savings. (13)

Healthcare costs vary significantly across countries, impacting the applicability of our findings in different economic contexts. For instance:

- **United Kingdom (NICE threshold: £20,000-£30,000/QALY)**: Cryotherapy does not meet the cost-effectiveness threshold but remains a viable lower-cost alternative
- **United States (Willingness-to-pay threshold: $50,000-$150,000/QALY)**: Cryotherapy could be considered cost-effective, particularly in lower-threshold scenarios
- **European Union**: Different reimbursement policies and cost structures impact the feasibility of cryotherapy, with some healthcare systems favoring surgical interventions due to superior QALYs
- **Low- and Middle-Income Countries (LMICs)**: Given budget constraints and limited access to high-cost surgical procedures, cryotherapy may provide a more feasible alternative, particularly if financial aid programs or lower-cost treatment protocols are implemented

This analysis has several important limitations that should be carefully considered when interpreting and applying its findings conducted in the United Kingdom, the study relies on NHS pricing models and UK-specific willingness-to-pay thresholds. Consequently, the results may not be applicable to healthcare systems characterized by different cost structures, clinical practices or economic valuations. The analysis employs EQ-5D-5L data collected at three key intervals: baseline, four weeks, and twelve months. While these data points offer a valuable glimpse into patient outcomes over a limited timeframe, they may not fully capture the complexities of long-term utility estimates. A potential limitation of EQ-5D-5L data is the profile values derived from the UK population. While the primary randomized controlled trial was conducted in the UK and using UK-specific values aligns with the study setting, it's important to acknowledge that different countries possess different value sets for the EQ-5D-5L. These variations could impact the generalizability of the QALY estimates and consequently the ICER results to other countries healthcare settings. Potential shifts in quality of life that could occur beyond the study's period may be overlooked. Although the Markov model tracks patient trajectories over a span of 20 years, its primary focus is on the costs and outcomes associated with initial treatment decisions. This specific lens may inadvertently neglect broader implications, such as indirect costs or societal impacts over extended time horizons. These limitations emphasize the need for cautious interpretation and application of the findings especially in different healthcare contexts or when considering long-term implications. The analysis may not account for several extreme scenarios that could impact its findings, including uncommon pathologies discovered through biopsies requiring specialized and costly treatments beyond standard care, severe life-threatening complications from the biopsy procedure itself affecting costs and quality of life and patients with multiple severe comorbidities or advanced disease stages that fall outside the typical profiles considered in the model.

Contribution

Aljira Fitya is responsible for populating Excel sheets, Sensitivity Analysis 4, drafting the initial report draft, contributing to the discussion

Anqi Zhao is responsible for inflation, QALYs_cryo, QALYs_surgery,ICER trial, decision tree, Tornado, ICER decision tree, Markov intervention, Markov control and ICER Markov in Excel sheets, drafting the abstract and CEA in the method part, and completing the figure and table section

Sijia Wei is responsible for QALYs for cryoablation input, sensitivity analyses (2,3,5) Excel sheets and decision trees, interpretation of sensitivity analyses result and report formatting.

Sonam Sharma is responsible for populating Cost cryoablation, Staff costs from PSSRU, QALY's, ICER Trial, CEP Graph in Excel sheets, drafting the introduction, aims of the study, discussion and proofreading the document.

Yinqi Qian is responsible for Excel sheets and the corresponding report sections on Cost of Biopsy, BNF, Stent, QALYs of Cryoablation, Costs, Cost-effectiveness, Sensitivity Analysis 1 (SA1 decision tree), Budget Impact Analysis and Markov Model. Also, in charge of the overall revision of the report and the Reference section.

Yuekun Liu is responsible for QALYs Excel sheets, Markov Excel sheets, Methods section and Markov section in report, contributing to Sensitivity Analysis discussion and other discussions

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